NICHOLAS LE ROUX

EMBEDDED SOFTWARE ENGINEER

C++ | Industrial protocols | IoT | Networking | APIs | Full-stack systems

Rotterdam, Netherlands / wmailnkwlr09@gmail.com / github.com/nkwleroux / linkedin.com/in/nicholas-lr

PROFILE

Systems-oriented engineer building software across hardware, industrial networks, protocol conversion, backend services, and connected applications. Professional focus is C++ embedded/systems work and modernizing established operational technology behind maintainable API boundaries.

EXPERIENCE

Embedded Software Engineer — SEP 2024 - PRESENT

Haia Consultancy

- Progressed from Software Engineering Intern to IFSF Technical Support and then full-time Embedded Software Engineer.
- Lead developer and system architect for ILAC, a two-engineer C++ bridge between OpenRetailing Price Pole REST APIs and installed IFSF LonWorks equipment.
- Authored IFSF Engineering Bulletin No. 27 and served as author/main contributor on Bulletin No. 26; contributed to an ILAC proof-of-concept demonstration to the IFSF Working Group.

Software Engineering Intern — AUG 2021 - JAN 2022

Axians

- Designed reusable C++ IoT components for sensor integration, data acquisition, wider system connectivity, and persistence.
- Framework was estimated to save about 40 engineering hours per month by reducing repeated setup work.

SELECTED ENGINEERING WORK

ILAC - IFSF LON API Converter

Lead Developer / System Architect / Team of 2. C++11, Oat++, REST/HTTP, IFSF, LonWorks/LonTalk, TP/FT-10, binary message parsing, acknowledgements, BCD/data conversion, and simulator/hardware validation.

ADDITIONAL PROJECT EVIDENCE

Connect 4
Java client/server game using TCP sockets and networked state.

Supermarket E-Shop
C# / WPF desktop webshop with TCP client/server communication and catalogue editor.

ESP32 Connected System
Embedded C / Espressif with Wi-Fi, HTTP, I2C/SMBus LCD, buzzer, clock, rotary and touch input.

Yonsei Quantitative Research
Python/Jupyter research replication using financial data, stock splits, and abnormal-return analysis.

CORE SKILLS

SYSTEMS
C / C++ (C++11), embedded software, concurrency, binary protocols, hardware integration

INDUSTRIAL
IFSF, LonWorks/LonTalk, TP/FT-10, REST/HTTP, TCP/IP, MQTT

BACKEND
C#, ASP.NET Core, Oat++, PostgreSQL, SQLite, MySQL/MariaDB

WEB / APP
TypeScript, JavaScript, React, Vite, React Native, Java, Kotlin, WPF

PLATFORM
Docker, Docker Compose, CMake, Git, Git LFS, WSL2, Azure, AWS, Prometheus, Grafana

HARDWARE
ESP32, STM32, Arduino, Raspberry Pi Pico, ATtiny85, EnOcean U60FT

INDUSTRY PUBLICATIONS

IFSF Engineering Bulletin No. 27
Sole author. Price Pole API Standard to IFSF LON conversion.

IFSF Engineering Bulletin No. 26
Author/main contributor. Renesas FT 5000 / FT 6050 end-of-life issue and migration considerations.

EDUCATION

Avans University of Applied Sciences
Computer Engineering, Netherlands | 2019 - 2025

Yonsei University
Computer Science study abroad, Seoul | Aug 2022 - Jul 2023 | Korean Language Institute Levels 1 + 2

AVAILABILITY

Interested in remote part-time work across full-stack, web, frontend/backend, cloud, embedded, systems, IoT, networking, and platform engineering.